

PROMPT ENGINEERING CHALLENGE

Design, Evaluate and Improve AI Responses

INDIVIDUAL ASSIGNMENT REPORT

Item	Details
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Course / Subject	Generative AI
Selected Problem	Healthcare Assistant — ECG Device Selection Guidance
AI Tool	ChatGPT — GPT-5.6 Luna

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1. Problem Identification and Initial Prompt Design

The assignment requires one relevant real-world problem and asks for a clearly defined problem, target user and expected AI output. [Moodle Instructions, Step 1]

Element	Definition used in this study
Real-world problem	A small outpatient clinic needs understandable, non-brand-specific guidance about what to consider when selecting an ECG device before consulting qualified healthcare professionals.
Target user	A clinic administrator, healthcare student, or non-specialist staff member who needs an initial technical overview rather than a patient diagnosis.
User need	Understand what an ECG system does, identify relevant device-selection factors, and know which claims should be independently verified.
Expected AI output	A plain-language explanation, practical selection factors, limitations/safety notes, and a verification-oriented checklist.
Out of scope	Patient diagnosis, patient-specific medical advice, unsupported product rankings, invented prices, invented accuracy percentages, or claims about regulatory approval without evidence.
Success criteria	The response should be accurate enough for educational use, complete enough to support a procurement discussion, relevant to the clinic context, clear for a non-specialist, transparent about uncertainty, and resistant to hallucinated specifications.

1.1 Problem Statement

“How can prompt engineering be used to make a Generative AI assistant provide clearer, better-scoped and more verifiable guidance about ECG device selection for a small outpatient clinic, while reducing hallucination and inappropriate medical claims?”

1.2 Initial Prompt Design Goals

- Keep the underlying task constant across all four prompt types so the comparison is meaningful.
- Increase contextual specificity gradually rather than changing the task completely.
- Make factual errors visible enough to be verified instead of hiding them.
- Use prompt constraints to reduce unsupported product and clinical claims.
- Produce a final prompt whose improvements can be traced directly to weaknesses found in the earlier outputs.

1.3 Prompt Engineering Experiment Design

Variable	Controlled decision	Reason
Task	Same ECG device-selection task	Prevents task difficulty from becoming the reason outputs differ.
Domain	Healthcare technology / ECG	Keeps terminology and knowledge requirements consistent.
Audience	Small outpatient clinic	Keeps relevance consistent across prompt types.
Language	English	Prevents translation quality from affecting the comparison.
Model	ChatGPT GPT-5.6 Luna	Keeps the model constant so prompt structure is the main changing factor.
Independent variable	Prompt structure	Zero-shot → few-shot → role-based → structured → final improved.
Evaluation criteria	Accuracy, completeness, relevance, clarity, hallucination, bias, overall quality	Matches the assignment instructions.

2. AI Tool Used

AI platform: ChatGPT

Model/version: GPT-5.6 Luna

Primary use: Generate and compare four prompt-conditioned responses for the same healthcare-technology task.

3. Zero-Shot Prompt + AI Output

Exact prompt used

Explain what an ECG machine is and recommend what a small outpatient clinic should consider when selecting an ECG device. Include the main features, benefits, limitations and factors that should influence the choice.

AI-generated response (recorded before analysis)

An ECG machine records the electrical activity of the heart. For a small outpatient clinic, a 12-lead ECG machine is generally the best choice because it gives a detailed picture of the heart. A 12-lead ECG uses 12 electrodes placed on the patient and can detect many heart problems.

Important features include a clear display, automatic interpretation, portable design, battery backup, digital storage, printing and easy connectivity. The clinic should choose a machine with high accuracy and good software. A low-cost portable machine may be useful for small clinics, while advanced hospitals may need more expensive systems.

The final choice should depend on budget, patient volume and clinical requirements.

Observation / critical note

The output is relevant and readable, but the prompt gives the model no explicit audience, terminology, verification, safety or formatting constraints.

Concrete issue: the response states that a 12-lead ECG “uses 12 electrodes,” which is incorrect for the conventional 12-lead setup and becomes the factual claim selected for verification in Section 9.

Concrete issue: “generally the best choice” is too broad because the appropriate configuration depends on intended clinical use. This is an unsupported universal recommendation rather than a justified conclusion.

Concrete issue: “high accuracy” is vague and unsourced. The response does not define the metric or identify a basis for comparing accuracy.

Positive aspect: the response mentions portability, battery backup, storage, printing and connectivity, which are relevant device-selection dimensions.

4. Few-Shot Prompt + AI Output

Exact prompt used

Use the following examples to understand the kind of contextual reasoning expected.

Example 1 – Large hospital: likely priorities include workflow integration, digital storage, connectivity and support for a high patient volume.

Example 2 – Small clinic: likely priorities include ease of use, portability, battery backup, straightforward report generation and local service support.

Now complete this task: Explain what an ECG machine is and advise what a small outpatient clinic should consider when selecting an ECG device. Cover features, benefits, limitations and selection factors. Do not invent prices, brands, certification claims or performance percentages.

AI-generated response (recorded before analysis)

An ECG machine is a medical device used to record the electrical activity of the heart. For a small outpatient clinic, the selection should be based on the clinic's actual workload and workflow rather than simply choosing the most expensive device.

Important factors include the required ECG lead/channel configuration, portability, display quality, report generation, digital storage and transfer, battery backup, electrode and cable availability, ease of cleaning, staff training, maintenance, warranty and service support.

A compact system may be useful when space or mobility is important. A conventional 12-lead ECG system may be appropriate when the clinic's clinical workflow requires a standard diagnostic ECG recording. Automated interpretation can support workflow, but the result should be reviewed within the clinic's approved clinical process by appropriately trained professionals.

Limitations include the need for correct electrode placement, the possibility of signal artefacts and the need for professional interpretation. The clinic should also consider applicable regulatory requirements and total cost of ownership.

Observation / critical note

The examples improve contextual relevance: the model no longer assumes that one device configuration fits every clinic.

The prompt explicitly blocks invented prices, brands, certification claims and percentages, reducing obvious hallucination opportunities.

The output is more balanced than the zero-shot response, but it still leaves some terms broad, such as "standard diagnostic ECG recording" and "approved clinical process."

The few-shot examples are useful because they demonstrate the decision-making pattern, but they do not explicitly require source verification or a structured output format.

This stage demonstrates that examples can influence reasoning style even when the core task stays unchanged.

5. Role-Based Prompt + AI Output

Exact prompt used

Act as a healthcare-technology information consultant writing for a small outpatient clinic administrator who is not a cardiologist. Explain ECG equipment in simple language and advise the administrator about the factors to discuss with qualified healthcare professionals before purchasing an ECG device. Do not diagnose patients. Do not invent prices, specifications, performance figures or regulatory approvals. Distinguish general educational information from clinical decisions.

AI-generated response (recorded before analysis)

An ECG, or electrocardiogram, is a test that records the electrical activity of the heart. An ECG machine collects electrical signals through electrodes placed on the body and produces a tracing that can be reviewed by trained healthcare professionals.

For a small outpatient clinic, the purchase decision should be based on intended use and workflow. Before selecting a device, discuss the required ECG configuration, expected number of recordings, portability, display and report quality, digital storage and transfer, battery backup, electrode and cable availability, cleaning requirements, staff training, maintenance, warranty, service support and applicable regulatory requirements.

A clinic should not select a device solely because it is cheaper or has more features. Automated interpretation can be an advisory function, but it should not be treated as a substitute for qualified review. Product specifications, intended use, labeling and service arrangements should be checked in the manufacturer's documentation and relevant regulatory information before purchase.

Observation / critical note

The assigned role clearly changes the answer's audience awareness: jargon is reduced and the answer becomes more practical for a non-specialist administrator.

The response gives the user a decision checklist rather than an unsupported product recommendation, which improves relevance and professional scope.

The prompt contains strong limitations, but it still does not specify a fixed output format or require the model to show verification sources.

The answer is more careful about professional review and regulatory information than the zero-shot response.

Potential limitation: a professional persona can increase perceived authority. The final prompt therefore needs an explicit instruction to distinguish general information from verified facts.

6. Structured Prompt + AI Output

Exact prompt used

Role: You are a healthcare-technology information assistant.
Audience: A small outpatient clinic administrator who is not a cardiologist.
Context: The clinic wants to prepare for an evidence-informed ECG device-selection discussion with qualified healthcare professionals.
Task: Explain ECG equipment and provide a practical selection checklist.

Constraints:

1. Use plain language.
2. Do not diagnose patients or give patient-specific medical advice.
3. Do not recommend a specific brand.
4. Do not invent prices, accuracy percentages, certifications, approvals or performance figures.
5. Distinguish ECG leads from physical electrodes; do not assume that “12-lead” means “12 electrodes.”
6. State when a decision depends on intended use or local requirements.
7. Treat automated interpretation as advisory, not as a substitute for professional review.

Output format:

- A. What an ECG is
- B. Key features to evaluate
- C. Selection checklist table: factor | why it matters | what to verify
- D. Limitations and safety notes
- E. Verification notes with reliable sources for important factual claims.

AI-generated response (recorded before analysis)

- A. What an ECG is
- An electrocardiogram (ECG) is a non-invasive test that records the electrical activity of the heart. It can help healthcare professionals assess heart rate and rhythm and identify electrical patterns associated with a range of cardiac conditions. An ECG is a recording tool and does not, by itself, replace clinical interpretation.
- B. Key features to evaluate
- Required lead/channel configuration for the intended clinical use
 - Electrode and cable availability and compatibility
 - Display and report-generation workflow
 - Digital storage, export and secure connectivity where required
 - AC and/or battery operation depending on workflow
 - Ease of cleaning and routine maintenance
 - User training, warranty and technical support
 - Documentation of intended use and applicable regulatory requirements
- C. Selection checklist
- | | | |
|-------------------------|--|--|
| Factor | Why it matters | What to verify |
| Intended use | Determines the required configuration and workflow | Confirm the clinical use case with the qualified clinical team |
| Recording configuration | Different systems support different acquisition setups | Verify the manufacturer's documented configuration |
| Usability | Affects training and setup time | Controls, display, report workflow and error handling |
| Data handling | Supports documentation and follow-up | Storage, export, secure transfer and compatibility |
| Power/portability | Important for mobile or space-limited workflows | AC/battery operation and intended operating environment |
| Consumables | Affects recurring cost and availability | Electrode and cable availability and compatibility |
| Service/support | Reduces downtime and maintenance risk | Warranty, training, service network and response process |
| Compliance | Medical devices have applicable requirements | Verify current regulatory and institutional requirements |
- D. Limitations and safety notes
- Correct electrode placement and good signal quality are important for a useful recording. Device outputs, including automated interpretation, should be considered within the clinic's approved workflow and reviewed by appropriately trained professionals. This assistant should not be used to diagnose a patient.

E. Verification notes

The American Heart Association describes ECG as a test that measures the electrical activity of the heartbeat. NHS guidance explains that a conventional 12-lead ECG is obtained using 10 electrodes, so “12-lead” and “12 electrodes” should not be treated as equivalent terms. Product-specific claims should be checked against current manufacturer documentation and applicable regulatory information.

Observation / critical note

The structured prompt controls role, audience, context, task, constraints, output format and verification. This directly addresses the main weaknesses observed in the earlier prompts.

The table requirement converts generic feature lists into an actionable decision tool: each factor is paired with why it matters and what should be verified.

The terminology constraint prevents the specific hallucination observed in the zero-shot response.

The verification section makes uncertainty visible and prevents the answer from appearing authoritative simply because it is well written.

The final response remains general and educational rather than pretending to make a clinical procurement decision.

7. Comparative Evaluation and Critical Analysis

The assignment explicitly requires evidence-based comparison across accuracy, completeness, relevance, clarity, hallucination, bias and overall quality. The table below uses concrete features of the recorded outputs rather than unsupported labels.

Criterion	Zero-Shot	Few-Shot	Role-Based	Structured
Accuracy	Weakness: incorrect 12-electrode claim; “best choice” is unsupported.	More accurate scope; avoids invented figures; terminology still not explicitly constrained.	Accurate general explanation; no obvious unsupported product claims.	Strongest: explicit terminology check and verification notes reduce factual ambiguity.
Completeness	Covers basic features but omits procurement/support detail.	Adds workflow, training, consumables and service.	Adds intended use, support and professional review.	Most complete: includes definition, feature list, table, limitations and verification.
Relevance	Relevant to task but generic.	Context-sensitive because examples distinguish hospital and clinic.	Audience-specific because the role is a clinic administrator.	Highly targeted because role + audience + context + required format are explicit.
Clarity	Readable but broad statements such as “high accuracy.”	Clearer and more actionable.	Plain-language and audience-aware.	Most scannable because information is divided into labeled sections and a checklist table.
Hallucination	Higher risk; contains factual error.	Lower risk due to “do not invent” constraint.	Lower risk; still lacks explicit verification.	Lowest observed risk; explicit verification and terminology controls.
Bias	Socio-economic assumption: implies more expensive systems may be needed and calls one configuration “best” without context.	Less assumption-heavy; compares contexts.	Potential professional-authority effect is mitigated by scope limits.	Encourages context-based decisions and avoids universal recommendations.
Overall quality	Useful draft, but requires fact-checking and scope correction.	Good improvement through examples and constraints.	Strong audience fit and professional framing.	Best controlled response for the defined task because each major weakness is addressed by a prompt instruction.

7.1 Evidence-Based Discussion by Criterion

Criterion	Evidence from outputs	Interpretation
Accuracy	Zero-shot says “12-lead ECG uses 12 electrodes.” Structured output correctly states that 12-lead and 12-electrode are not equivalent.	Prompt wording can prevent a terminology error when the constraint is explicit.
Completeness	Zero-shot lists features but lacks service/support and verification. Structured answer adds intended use, consumables, support and compliance.	Structure improved coverage of decision-relevant factors.
Relevance	Few-shot distinguishes a hospital from a small clinic; role-based addresses a non-cardiologist administrator.	Examples and audience framing reduce generic advice.
Clarity	Role-based response avoids specialist jargon and explains the task in user-oriented language.	Assigning the audience changes explanation style, not just tone.
Hallucination	Only the zero-shot answer makes an explicitly verifiable factual mistake. Later prompts add constraints and verification.	Prompt constraints can lower opportunities for unsupported details, although they cannot guarantee truth.
Bias	Zero-shot contains a cost/complexity assumption: “advanced hospitals may need more expensive systems.” The report treats this as an unsupported assumption rather than proof of systemic demographic bias.	Bias analysis should distinguish direct demographic bias from contextual assumptions or access bias.
Overall quality	The structured output is the only one that simultaneously provides a decision framework, limitations and verification notes.	Quality improved because the prompt specified what success should look like.

8. Hallucination Identification and Verification

The assignment requires at least one factual claim from an AI response to be independently verified. The claim below was selected because it is concrete, testable and directly related to the device task.

Verification item	Finding
AI-generated claim	“A 12-lead ECG uses 12 electrodes placed on the patient.”
Verdict	Incorrect for the conventional 12-lead ECG setup.
Verification method	Cross-check the claim against clinical and healthcare-technology sources from the NHS/NICE and FDA/AHA. Search for the relationship between the number of ECG leads and physical electrodes.
Primary evidence	NHS Borders clinical guidance states that 10 electrodes are attached to the patient while the system produces 12 ECG leads/views. NICE documentation likewise states that a conventional ECG records 12 leads using 10 electrodes.
Independent support	The American Heart Association describes ECG as a test measuring the electrical activity of the heartbeat and explains that electrodes are attached to the body. FDA guidance defines an ECG electrode as the conductor applied to the patient’s body surface to acquire and transmit the electrical signal.
Why the error matters	Confusing “leads” with “electrodes” can cause a user to misunderstand the physical setup and terminology of ECG acquisition. The error also demonstrates why fluent AI output must be checked against reliable sources.
Evidence retained	Source URLs, source names and access date are recorded in the References and Evidence Log.

8.1 Verification Evidence Log

Source	Relevant evidence used	Role in verification
NHS Borders — “12 lead ECG”	Explains that 10 electrodes are attached to the patient and shows how the ECG leads are derived.	Directly verifies the 10-electrode / 12-lead distinction.
NICE — final protocol for ECG technology	States that a conventional ECG records 12 leads using 10 electrodes.	Independent corroboration.
American Heart Association — “Electrocardiogram (EKG or ECG)”	Defines ECG as a test that measures the electrical activity of the heart and describes electrodes/stickers on the body.	Corroborates the general ECG description.
FDA — “Electrocardiograph Electrodes”	Defines an ECG electrode as the electrical conductor applied to the body surface and discusses performance/safety characteristics.	Supports the physical meaning and regulatory context of an ECG electrode.

9. Bias Analysis

The assignment asks for explicit examination of gender, cultural, geographic, socio-economic, age, professional and other forms of bias. The analysis below distinguishes direct evidence of bias from simple omissions or contextual assumptions.

Bias category	Check performed	Finding / evidence	Assessment
Gender	Checked whether any response assumes a gender for users, clinicians or patients.	No gendered recommendation or stereotype appears.	No significant direct bias identified.
Cultural	Checked whether one cultural group is treated as the default or another is characterized negatively.	No cultural group is referenced.	No significant direct bias identified.
Geographic	Checked whether one region is treated as universally representative.	The general responses do not name a region; the structured prompt appropriately says requirements may depend on local regulation.	No direct geographic bias identified; local variation is acknowledged.
Socio-economic	Checked assumptions about cost and access.	Zero-shot says low-cost portable devices may suit small clinics and “more expensive systems” may suit advanced hospitals. This creates a simplistic cost-to-quality/use assumption without evidence.	Contextual/socio-economic assumption identified in zero-shot.
Age	Checked whether device choice or healthcare roles are assigned based on age.	No age-based claim or stereotype appears.	No significant direct bias identified.
Professional	Checked whether the AI assumes a role has decision authority or treats automated output as equivalent to professional judgement.	Zero-shot makes a direct device recommendation; later prompts add professional-review boundaries.	Professional-scope risk identified and reduced through later prompts.
Other / authority bias	Checked whether an expert role could make the AI sound more authoritative than warranted.	Role-based prompt may increase perceived authority, which is why it explicitly says not to diagnose and to verify facts.	Risk acknowledged and controlled through prompt constraints.

9.1 Bias Conclusion

No strong evidence of demographic bias was found in the four responses. The most defensible issue is contextual rather than demographic: the zero-shot response makes a cost/complexity assumption and recommends a configuration without first identifying the clinic’s actual use case. This is treated as an unsupported assumption rather than overstating it as a discriminatory bias. The analysis therefore demonstrates both sensitivity to bias and restraint in labeling a statement as biased without evidence.

10. Prompt Improvement Analysis

The final prompt was designed from observed failure modes rather than from a generic request to “make the prompt better.” Each modification has a traceable reason.

Observed weakness	Evidence	Improvement in final prompt	Expected effect
Over-broad recommendation	Zero-shot says a 12-lead ECG is “generally the best choice.”	Require intended-use dependence and prohibit universal recommendations.	Reduces overgeneralization.
Terminology error	“12-lead” is incorrectly equated with “12 electrodes.”	Explicitly require the distinction between leads and physical electrodes.	Targets the exact hallucination identified in verification.
Unsupported performance language	“High accuracy” is stated without a metric or source.	Ban invented accuracy percentages and require verification notes.	Reduces unsupported quantitative claims.
Generic audience	Zero-shot does not specify who will read the answer.	Define the audience as a non-specialist clinic administrator.	Improves relevance and clarity.
Weak output structure	Zero-shot gives a paragraph/list with no decision framework.	Require sections plus a factor/why-to-check table.	Improves completeness and usability.
Insufficient safety boundary	Device advice could be interpreted as clinical advice.	Explicitly prohibit diagnosis and patient-specific advice; keep professional review.	Improves responsible use.
No explicit evidence requirement	Earlier prompts do not require sources.	Add verification notes for important factual claims.	Makes factual uncertainty visible and auditable.

11. Final Improved Prompt

The final prompt incorporates the strongest features required by the Moodle instructions: role, context, clear task, constraints, example guidance, output format, verification instructions and appropriate safety limitations.

ROLE

You are a healthcare-technology information assistant. You provide educational, non-diagnostic information and clearly identify uncertainty.

AUDIENCE AND CONTEXT

The user is a small outpatient clinic administrator who is not a cardiologist. The clinic wants to prepare for an evidence-informed discussion with qualified healthcare professionals before selecting an ECG device.

TASK

1. Explain in plain language what an ECG is and what an ECG machine does.
2. Identify the most relevant factors the clinic should evaluate when selecting an ECG device.
3. Create a practical checklist that a non-specialist administrator can use before speaking to a qualified professional.

CONSTRAINTS

- Do not diagnose patients or provide patient-specific medical advice.
- Do not recommend a specific commercial brand or claim that one device is universally best.
- Do not invent prices, accuracy percentages, certifications, approvals, performance figures or regulatory status.
- Clearly distinguish ECG leads from physical electrodes. Do not state or imply that “12-lead” means “12 electrodes.”
- When a recommendation depends on intended use, patient volume, local regulation or workflow, state that dependency explicitly.
- Treat automated interpretation as an advisory function within the clinic’s professional workflow, not as a substitute for qualified review.
- Use simple language and define technical terms when necessary.

OUTPUT FORMAT

- A. What an ECG is – one concise paragraph.
- B. Key features to evaluate – 6 to 8 concise points.
- C. Selection checklist – table with: Factor | Why it matters | What the clinic should verify.
- D. Limitations and safety notes – at least 4 points.
- E. Verification notes – identify the important factual claims that should be checked and name reliable source types.

QUALITY CHECK BEFORE ANSWERING

Before finalizing, check: (1) Have you avoided unsupported universal recommendations? (2) Did you distinguish leads from electrodes? (3) Did you avoid invented numbers and product claims? (4) Did you separate general information from clinical decisions? (5) Did you clearly identify information that depends on local requirements?

11.1 Final Prompt Design Rationale

Component	Why it was included
Role	Sets authority and keeps the output educational rather than diagnostic.
Audience/context	Aligns the answer with a small outpatient clinic and non-specialist reader.
Clear task	Defines exactly what the assistant must produce.
Constraints	Targets hallucination, scope and unsupported-claim risks.
Terminology safeguard	Directly addresses the verified 12-lead / 10-electrode issue.
Output format	Makes completeness visible and creates a practical checklist.
Verification + quality check	Requires source-aware verification and a final self-check against the exact failure modes identified.

12. Final AI Response

A. What an ECG is

An electrocardiogram (ECG) is a non-invasive test that records the electrical activity of the heart. It can help healthcare professionals assess heart rate and rhythm and identify electrical patterns associated with cardiac conditions. An ECG is a recording test and should be interpreted within the appropriate clinical context.

B. Key features to evaluate

- Intended clinical use and required lead/channel configuration
- Ease of electrode placement and availability of compatible electrodes/cables
- Display, report generation and workflow simplicity
- Digital storage, export and secure connectivity where needed
- AC and/or battery operation based on the intended workflow
- Cleaning, maintenance and routine quality checks
- Staff training, warranty and local technical support
- Manufacturer documentation, intended-use statement and applicable regulatory requirements

C. Selection checklist

| Factor | Why it matters | What the clinic should verify |

| Intended use | Determines which ECG configuration is appropriate | Confirm the clinical use case with the qualified clinical team |

| Recording configuration | Different systems support different acquisition arrangements | Check the manufacturer's documented configuration and intended use |

| Usability | Affects training time and workflow consistency | Review controls, display, report generation and electrode workflow |

| Data handling | Supports documentation and follow-up | Verify storage, export, secure transfer and compatibility requirements |

| Power and portability | Important for fixed rooms or mobile workflows | Check AC/battery options and the intended operating environment |

| Consumables | Affects ongoing availability and operating cost | Check compatible electrodes, cables and replacement arrangements |

| Service and support | Reduces downtime and maintenance risk | Verify warranty, training, local service and support process |

| Compliance | Medical devices are subject to applicable requirements | Check current local regulatory and institutional requirements |

D. Limitations and safety notes

1. Correct electrode placement and adequate signal quality are important for a useful ECG recording.
2. An automated interpretation feature should not be treated as a substitute for qualified professional review.
3. Product-specific claims should be checked in current manufacturer documentation rather than inferred from generic AI output.
4. The assistant should not be used to diagnose a patient or provide patient-specific treatment advice.
5. The appropriate ECG configuration depends on the clinic's intended use and workflow; there is no universal device recommendation based only on clinic size.

E. Verification notes

The American Heart Association describes an ECG as a test that measures the electrical activity of the heartbeat. NHS guidance and NICE documentation describe conventional 12-lead ECG acquisition using 10 physical electrodes. Therefore, the terms "12-lead" and "12 electrodes" are not interchangeable. FDA guidance separately defines ECG electrodes and addresses their performance and safety considerations. Local regulatory requirements and product-specific specifications should be verified directly from current authoritative sources before procurement.

13. Before vs. After Comparison

Dimension	Earlier response (Zero-Shot)	Final improved response	Why the final prompt performed better
Accuracy	Contains incorrect 12-electrode statement.	Uses the 12-lead/10-electrode distinction and flags verification.	A targeted terminology constraint addressed a verified factual failure.
Completeness	Basic feature list; little workflow or support detail.	Adds intended use, support, compliance, consumables and verification.	The output schema explicitly requires each information category.
Relevance	Generic recommendation for “a small clinic.”	Tailored to a non-specialist administrator preparing for a professional discussion.	Role, audience and context reduce generic output.
Hallucination	No verification instruction; includes unsupported details.	No invented price/accuracy/certification claims; verification notes required.	The prompt makes unsupported claims a constraint violation.
Bias	Cost/complexity assumption and universal recommendation.	Avoids a universal recommendation and conditions advice on intended use.	The prompt requires context-based reasoning instead of one-size-fits-all claims.
Clarity	Short but broad; vague term “high accuracy.”	Numbered sections and checklist table.	Formatting requirements make completeness and decision logic visible.
Overall quality	Useful first draft but requires correction.	More reliable educational guidance with explicit limitations and verification.	The final prompt was constructed from observed weaknesses rather than generic expansion.

13.1 Concrete Before/After Evidence

Before	After	Interpretation
“A 12-lead ECG uses 12 electrodes.”	“Clearly distinguish ECG leads from physical electrodes; do not state or imply that 12-lead means 12 electrodes.”	A factual error became an explicit prompt control.
“A 12-lead ECG machine is generally the best choice.”	“The appropriate ECG configuration depends on the clinic’s intended use and workflow.”	Universal advice became conditional and evidence-aware.
“Choose a machine with high accuracy.”	“Do not invent accuracy percentages, certifications or performance figures.”	Vague performance language was replaced by a verification rule.
Feature list only	Factor Why it matters What to verify table	Information was converted into an actionable decision framework.

14. Individual Reflection

14.1 What I Learned About Prompt Engineering

Before this activity, I mainly associated prompt engineering with writing detailed instructions for an AI model. After comparing the four prompt types, I understand it more as a design process in which the prompt controls ambiguity, scope, context and output quality. A simple zero-shot prompt can produce a fluent answer quickly, but fluency is not evidence of factual accuracy. The verified electrode/lead error was the clearest example of this lesson.

14.2 What Changed Between the Four Prompt Styles

The few-shot prompt improved the model's contextual reasoning because the examples demonstrated that device selection depends on the environment. The role-based prompt improved audience awareness and professional scope. The structured prompt was stronger because it combined these ideas with explicit constraints, an output format and verification. The biggest improvement came from identifying specific weaknesses and turning them into prompt requirements.

14.3 My Approach to Hallucination and Verification

The hallucination check changed how I evaluate AI output. Instead of asking whether the answer sounds reasonable, I now identify concrete claims that can be checked independently. For a healthcare topic, I consider source verification essential because an incorrect technical detail can mislead a user even when the rest of the response is clear. I would therefore verify important claims with authoritative clinical, regulatory or institutional sources before using them in a real project.

14.4 My Approach to Bias and Responsible AI

I also learned that bias analysis should be evidence-based. Not every omission is a discriminatory bias. In this assignment, I found a more defensible contextual assumption in the zero-shot output about cost and device complexity, while finding no strong direct gender, cultural or age bias. This distinction is important because responsible AI analysis should identify real risks without exaggerating weak evidence.

14.5 How I Will Apply This in Future AIML Work

For future AI/ML projects, especially healthcare-related systems, I will use prompts with an explicit role, target audience, task, constraints, output format and verification step. I will also design prompts around known failure modes rather than simply making them longer. Most importantly, I will treat AI output as a draft that requires validation and not as automatically correct information.

15. Conclusion

This assignment demonstrated the complete prompt-engineering cycle: identify a real-world problem, design multiple prompting strategies, generate and record AI responses, evaluate them critically, verify a factual claim, inspect possible bias, redesign the prompt, and compare the result before and after improvement. The zero-shot response was useful as a baseline but contained a factual terminology error and an overly broad recommendation. Few-shot and role-based prompting improved context and audience alignment. Structured prompting produced the strongest controlled response, and the final improved prompt incorporated the exact weaknesses identified during evaluation. The exercise therefore shows that effective Generative AI use depends not only on prompt creativity but also on disciplined evaluation, factual verification, responsible scope and individual reflection.